

The 0/0 Atlas

A Complete Catalog of Indeterminate Form in Mathematics

The L.O.R.E. Collaboration

Puronbo Laboratory

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THE 0/0 ATLAS

A Complete Classification of Indeterminate Forms Across Mathematics

Authors: The L.O.R.E. Collaboration

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Abstract

We present a complete classification of 76 entries for the indeterminate form 0/0 across seventeen branches of mathematics: 64 verified instances with removable values (63 from the missing-experiment sweep plus the Landau-function calibration #76 -- removable value 1, $\log g(n)/\sqrt{n \log n} \rightarrow 1$, Landau 1903 -- added 2026-09-22, treated in §2.20), 6 constructive-program entries targeting open conjectures, named constants, and a manufactured/companion control (twin-prime and strong Goldbach, both removable value 1; the prime-reciprocal constant $B?$, removable value 0.2614972...; the Euler-totient density, removable value $3/\pi^2 = 0.30396355...$; the fixed-character bounded-box control, removable value 0; the Mertens product-gap complement, removable value 0 -- elementary, Mertens 1874), and 6 anti-class entries (Pólya, Mertens, the Dirichlet divisor-error, the Gauss circle-error, the character-walk sup-ratio, and the Chebyshev-walk delta) proving that 0/0 forms without removable value exist -- the class the corpus previously counted as empty. Each instance is classified by (i) the branch, (ii) the mechanism (Probe, Index, Vanishing Rate, Critical Phenomenon, or Conservation), (iii) the exact 0/0 form, (iv) the removable value, (v) the theorem it encodes, and (vi) the computational verification. We provide cross-reference tables, a decision tree for classifying new instances, and a catalog of open problems.

Part I: Definitions

1.1 The indeterminate form

Definition. An expression f/g is an indeterminate form of type $0/0$ at $x?$ if:

1. f and g are both defined in a punctured neighborhood of $x?$
2. $\lim_{x \rightarrow x?} f(x) = 0$ and $\lim_{x \rightarrow x?} g(x) = 0$
3. The limit $\lim_{x \rightarrow x?} f(x)/g(x)$ exists and is finite

The removable value is the limit.

Remark. Condition 3 is not automatic -- the limit might not exist (e.g., $f(x)/g(x) = \sin(1/x)$ as $x \rightarrow 0$). When it does exist, the singularity is removable.

1.2 The five mechanisms

Mechanism I (Probe). $f/g = 1$ where defined, but f and g both vanish at isolated points. The removable value at each zero tests whether f and g are "the same."

Mechanism II (Index). The $0/0$ arises from an integral of the form $\int ? (\text{something})/(\text{something else})$ around a zero. The removable value is an integer -- a winding number, index, or multiplicity.

Mechanism III (Vanishing Rate). A quantity $h(t)$ vanishes at $t = 0$. The ratio $h(t)/t^n$ is $0/0$. The removable value is the leading Taylor coefficient.

Mechanism IV (Critical Phenomenon). At a phase transition, two divergent or vanishing quantities form a $0/0$ or $0 \times \text{inf}$. The removable value is a critical amplitude.

Mechanism V (Conservation). A conserved quantity is the ratio of two quantities that both vanish when the symmetry is broken. The removable value is the conserved quantity.

Part II: The Complete Catalog

2.1 Number theory

#	Name		0/0 Form		Point		Removable Value		Mechanism		Verified	
1	Riemann zeta		zeta(s)	/	zeta(1?s)	zero rho		chi(rho)	= 1 iff Re(rho) = 1/2	Probe	Yes	
2	GRH Dirichlet		L(s,chi)	/	L(1?s,chi?)	zero rho		epsilon(chi)	= 1	Probe	Yes	
3	BSD		L(s,E)/(s?1)^r		s=1		Leading coefficient		Probe	Yes		
4	abc conjecture		log(c)/log(rad(abc))		(1,0,1)		1		Vanishing Rate		Yes	
5	Fermat little		(a^{p?1}-1)/(a-1)		a=1		p?1		Vanishing Rate		Yes	
6	Euler product		Prod(1-p^{?s})/zeta(s)		s=1		1		Probe		Yes	
7	Weil explicit		?zeta'/zeta vs Sum 1/p^s		s=1		Sum log(p)/(p^s-1)		Prime identity		Probe	Yes
8	Zeta FE		zeta(0) via FE		s=0		?1/2		Vanishing Rate		Yes	
9	PNT		pi(x).log(x)/x		x->inf		1		Vanishing Rate		Yes	
10	Khintchine	q.	x?p/q		/psi(q)		convergent	1/sqrt5 (golden ratio)	Vanishing Rate		Yes	
11	Möbius		(s?1)/zeta(s)		s=1		1		Probe		Yes	

2.2 Complex analysis

#	Name	0/0 Form	Point	Removable Value	Mechanism	Verified
12	Argument principle	$f'(z)/f(z)$	zero rho	multiplicity k	Index	Yes

13	Cauchy integral	$f(z)/(z^?a)$	zero a	$f'(a)$	Vanishing Rate	Yes
14	Picard little	$f(z)/z^?k$	zero of order k	$f^{(k)}(0)/k!$	Vanishing Rate	Yes
15	Taylor remainder	$R_n(x)/(x^?a)^{n+1}$	$x=a$	$f^{(n+1)}(a)/(n+1)!$	Vanishing Rate	Yes
16	FTA	$f(z)/(z^?z_0)^?k$	root z_0	$g(z_0)$	Vanishing Rate	Yes
17	Stirling	$(n!/S^?1) \cdot n$	$n \rightarrow \infty$	$1/12$	Vanishing Rate	Yes
18	Wallis product	$\text{Prod}(2n)^2/((2n^?1)(2n^?1))$	$n \rightarrow \infty$	$\pi/2$	Vanishing Rate	Yes
19	Cesàro	$(1^?1+1^?1+...) \text{ mean}$	$n \rightarrow \infty$	$1/2$	Vanishing Rate	Yes

2.3 Algebraic topology and geometry

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
20	Poincaré-Hopf	$\chi(M) = \sum \text{index}(v_i)$	$\sum \text{index}(v_i)$	zero of V	Index (integer)	Yes
21	Atiyah-Singer	$\dim \ker(D) - \dim \text{coker}(D)$	Dirac operator	topological index	Index	Yes
22	Gauss-Bonnet	$\int K dA / (2\pi \chi)$	surface	1	Index	Yes
23	Riemann-Roch	$\chi(D) = \deg(D) - g + 1$	divisor D	$\deg(D) - g + 1$	Index	Yes
24	Weyl law	$N(\lambda)/\lambda^{d/2} \rightarrow 0$	$\lambda \rightarrow \infty$	Weyl constant	Index	Yes
25	Selberg trace	$\text{Tr}(e^{it\Delta})$	$t \rightarrow \infty$	1 (zero mode)	Index	Yes
26	Lefschetz	$\sum (-1)^i \text{ind}(p_i)$	fixed points	Euler characteristic	Index	Yes
27	Morse theory	$f(x)/Q(x)$ (Hessian critical)	critical point	± 1 (Morse index)	Index	Yes
28	Sard theorem	critical values measure 0	critical set	0	Index	Yes
29	Stokes/de Rham	$\int_M d\omega = \int_{\partial M} \omega$	boundary	0	Index	Yes
30	Green's function	$G(x, x)$	diagonal	eigenfunction reciprocity	Index	Yes
31	Euler-Maclaurin	$x/(e^x - 1)$	$x=0$	1 (B_0)	Vanishing Rate	Yes

2.4 Analysis and approximation

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
32	Central limit	$(\phi(t)^2)^{1/t^2}$	$t=0$	$\sigma^2/2$	Vanishing Rate	Yes
33	Rayleigh quotient	$t(Ax \cdot x)/(x \cdot x)$	$x=0$	eigenvalue	Vanishing Rate	Yes
34	Banach fixed point	$T(T(x) - x)/(x - x^*)$	$x=x^*$	$T'(x^*) - 1$	Vanishing Rate	Yes
35	Brouwer fixed point	$(f(x) - x)/(x - x^*)$	$x=x^*$	$f'(x^*) - 1$	Vanishing Rate	Yes
36	Fourier uncertainty	$\Delta f \Delta x = 1/4$	$\Delta f \Delta x$	uncertainty bound	Vanishing Rate	Yes
37	Poisson summation	$\sum f(n)/\sum f^*(n)$	trivial	1	Probe	Yes
38	Saddle point	$g'(x)/(x - x^*)$	$x=x^*$	$g''(x^*)$	Vanishing Rate	Yes
39	Laplace method	$\ln(n) \cdot \sqrt{n}$	$n \rightarrow \infty$	$\sqrt{\pi}$	Vanishing Rate	Yes
40	Noether-Landau	dF/ds	$s=0$	Landau coefficient	Conservation	Yes

2.5 Mathematical physics

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
41	Ising model	χ/T_c	T_c	critical amplitude	Critical	Yes
42	Spectral gap	$\Delta(L) \cdot L^z$	$h \rightarrow 1$	$C \sim \pi$	Critical	Yes
43	Lorenz attractor	$\log(\delta)/t$	$t \rightarrow 0$	$\lambda_1 \sim 0.91$	Critical	Yes
44	Wigner semicircle	$\rho(\lambda)/\sqrt{4 - \lambda^2}$	$\lambda = \pm 2$	$1/(2\pi)$	Critical	Yes
45	Semicircle	$N(E)/E$	$E \rightarrow 0$	$1/\pi$	Vanishing Rate	Yes

2.6 Information theory and statistics

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
46	Shannon entropy	$p \cdot \log(p)$	$p \rightarrow 0$	0	Critical	Yes
47	Boltzmann entropy	$S/\ln(W)$	$W=1$	1	Critical	Yes
48	Bayes theorem	$P(H D)$	$P(D) \rightarrow 0$	$P(H)$ (prior)	Conservation	Yes
49	Fourier uncertainty	$\sigma_x \sigma_p$	$\sigma_p \rightarrow 0$	constant (bound)	Vanishing Rate	Yes

2.7 Optimization and control

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
50	Gradient descent	$\Delta\theta/\eta$	$\eta \rightarrow 0$	??L	Conservation	Yes
51	KKT conditions	$\mu_i/g_i(x^*)$	active constraint	shadow price lam	Conservation	Yes
52	Schanuel	$e^{\alpha_1}/e^{\alpha_2} = \alpha_1 - \alpha_2$	$\alpha_1 = \alpha_2$		Conservation	Yes
53	Noether theorem	$dL/d\epsilon$	$\epsilon = 0$	conserved quantity	Conservation	Yes

2.8 Algebra

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
54	Pythagorean theorem	$a^2 + b^2 = c^2$ at right angle	$\theta = \pi/2$	0 (with $c = \text{hypotenuse}$)	Conservation	Yes

2.9 Dynamical systems

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
55	Poincaré recurrence	$\epsilon \cdot \tau(\epsilon)$	$\epsilon \rightarrow 0$	constant	Critical	Yes

2.10 Missing-experiment sweep (2026-09-20): eight new branches

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
56	Galois discriminant	$\text{Disc}(t)/t^{\{m\}}$	repeated root, $t \rightarrow 0$	$C = 32$ (double), $27a$ (triple)	Vanishing Rate	Yes
57	Bott periodicity	$K \sim (S^n)/\dots$ degenerate spectrum	$n \rightarrow 0$ (complex periodic)	$1/2$ (degenerate spectrum)	Vanishing Rate	Yes
58	Ricci flow neck pinch	$R \rightarrow A$	pinch ($R \rightarrow \infty$, $A \rightarrow 0$)	0 (Gauss-Bonnet)	Conservation	Yes
59	Natural transformant	$n(P)/\text{tr}(P?)$	degenerate object	1 (if $n = 1$ for $k=1$, 0 for $k=2$)	Vanishing Rate	Yes
60	Generating function	$\ln \rho \sim n^\alpha$	radius of convergence	$1/4$ (if $\rho = 1/4$, $1/2$ for $1/2$)	Vanishing Rate	Yes
61	Birkhoff average	(averages of collapsed points)	point 1 (generic) / 0 (exceptional)	1 (generic) / 0 (exceptional)	Vanishing Rate	Yes
62	Spectral density	$\rho \sim (2E)^{1/2}$	band edge $E \rightarrow 2$	$1/(2\pi)$, exponent $1/2$	Vanishing Rate	Yes
63	Néron-Tate height	$h(P)/c_s$	torsion point	$C \sim 3/8$ (log channel)	Vanishing Rate	Yes

2.11 Open-target construction (2026-09-20): conjectures and the anti-class

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
64	Pólya / Liouville	$(\sum_{n \leq x} \Lambda(n) - x) / \sqrt{x}$	Signum (reduced)	NONE -- the limit does not exist	Does class exist? (Yes)	Conjecture (Littlewood); conjecture
65	Twin-prime density	$\pi_2(x)/(2C \cdot \text{Li}(x)) \rightarrow \infty$		1 (Hardy-Littlewood)	Vanishing Rate	Yes
66	Goldbach representation	$\sum_{n \leq x} \Lambda(n) \log n \rightarrow \infty$		1 (strong Goldbach)	Pr, OPEN	Yes

2.12 The anti-class matures and a constant is named (2026-09-20)

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
67	Mertens function	$M(x)/\sqrt{x}$	NONE -- the limit does not exist	1 (if $M(x) \sim \sqrt{x}$, 0 for $M(x) \sim x$)	Does class exist? (Yes)	Conjecture (Littlewood); conjecture
68	Prime-reciprocal	$\sum_{p \leq x} 1/p$	$x \rightarrow \infty$	$0.2614972128\dots$	Pr, OPEN	Yes, named constant

2.13 The lattice-point family enters the anti-class (2026-09-21)

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
69	Dirichlet divisor error	$\Delta(N)/N^{1/4}$	$N \rightarrow \infty$	NONE -- the limit does not exist	Does class exist? (Yes)	Conjecture (Littlewood); conjecture

2.14 The Hardy pair completes: the circle member (2026-09-21)

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
70	Gauss circle error	$P(N)/\sqrt{N}$	$N \rightarrow \infty$	NONE -- the limit does not exist	Does class exist? (Yes)	Conjecture (Littlewood); conjecture

2.15 The flat-decay removable anchor (2026-09-21)

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
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71	Euler-totient density $\sum_{k \leq x} \phi(k)/x^2$, $\Phi = \lim_{x \rightarrow \infty} \sum_{k \leq x} \phi(k)/x^2 = 0.3039635599140613243$; Dirichlet 1849, proved: $\Phi = 3/\pi^2$
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2.16 The bounded-box control: removable under maximal raw oscillation (2026-09-21)

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
72	Character pair (S) and (Z) in Sigma_1 (C-Poly(Vinyl Chloride)) improved the pair (S) and (Z) in the manufacturing process					

2.17 The anti-class enters q-space (2026-09-21)

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
73	Character-Reliability	Supra_50 (ANTI-OSS)	S_qq_Sigma (muCH)	(muONE -- the liver does)	Agarichis (MORONET)	ONLY: Paley 19

2.18 The Chebyshev walk completes the sqrt n trinity (2026-09-21)

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
74	Chebyshev walk	$\delta_{\text{Chebyshev}} = \frac{1}{2} \left(\frac{1}{\sqrt{1 - \frac{1}{4n}} + \frac{1}{4n}} \right)$	$\frac{1}{2} \left(\frac{1}{\sqrt{1 - \frac{1}{4n}} + \frac{1}{4n}} \right)$	$\frac{1}{2} \left(\frac{1}{\sqrt{1 - \frac{1}{4n}} + \frac{1}{4n}} \right)$	Montgomery-Vaughan	ADDITIONALLY: Montgomery-Vaughan

2.19 The Mertens product gap completes the n-space complement (2026-09-21)

[illegible]

2.20 The Landau-function calibration enters the named-value 1 set (2026-09-22)

#	Name	O/O Form	Point	Removable Value	Mechanism	Verified
76	Landau function / pinog n)	(RE) Wave Bldg. Rule C-0001 or -C00Azer_0 up to 99999 (-x(x)(y))(-z(z)(t))P(0)g(y)s(-S)for n=0				referee

Part III: Cross-References

3.1 By removable value type

Removable Value Type	Experiments	Count
Integer	Poincaré-Hopf, Argument principle, Atiyah-Singer, Morse, Fermat little, FTA	1
Rational (simple)	Euler-Maclaurin (1), Cesàro (1/2), Taylor (1/n!)	2
Real (transcendental)	Zeta FE (?1/2), CLT (?sigma²/2), Wigner (1/(2pi)), Laplace (sqrt(pi))	3
1 (identity)	Riemann zeta, GRH, Euler product, Poisson, Schanuel, Stokes, Green, Gauss-Bonnet	4
0	Shannon, Boltzmann (S/ln(W)), Sard, Morse saddle, character partial sums (#72)	5
Function of parameters	BSD (rank+Sha), Ising (C), Spectral gap (1/C), Lorenz (lambda), Khintchine (1/sqrt(C))	6
New (56-63)	Galois (32/27), Bott (4, then 1), Ricci (8pi/3), Natural (1), Generating (1/sqrt(pi))	7
Real (named constant)	Prime-reciprocal constant B? (#68), Euler-totient density 3/pi² (#71)	8
Nonexistent (not removable)	Pólya F(n)/sqrt(n) (#64), Mertens M(n)/sqrt(n) (#67), Chebyshev delta(x)/sqrt(x) (#74)	9

3.2 By mathematical domain

Domain	Experiments	Mechanisms Used
Number theory	1-11	Probe (6), Vanishing Rate (5)
Complex analysis	12-19	Index (2), Vanishing Rate (6)
Algebraic topology/geometry	20-31	Index (11), Vanishing Rate (1)
Analysis/approximation	32-40	Vanishing Rate (7), Probe (1), Conservation (1)
Mathematical physics	41-45	Critical (3), Vanishing Rate (2)
Information theory	46-49	Critical (2), Conservation (1), Vanishing Rate (1)
Optimization	50-53	Conservation (4)
Algebra	54	Conservation (1)

Dynamical systems	55	Critical (1)
New branches (sweep)	56-63	Probe (2), Index (1), Vanishing Rate (4), Conservation (1)
Open-target construction	64-76	Vanishing Rate (4), Probe (3), Anti-class (6)

3.3 By mechanism

Mechanism	Experiments	Count	Character
Probe	1, 2, 3, 6, 7, 37, 59, 61, 66, 68, 172	12	Tests identity of two objects
Index	12, 20-31, 57	14	Extracts integer (winding, multiplicity)
Vanishing Rate	4, 5, 8, 9, 10, 11, 15-19, 32-36, 38-40, 45, 49, 56, 60, 62, 63, 65, 71	36	Calculating Taylor coefficient
Critical	41-43, 46, 47, 55	6	Phase transition / critical amplitude
Conservation	48, 50-54, 58	7	Conserved quantity from symmetry
(Anti-class)	64, 67, 69, 70, 73, 74 (Pólya, Mertens, divisor-error, circle-error, outside the five mechanisms: refutes "all are re	0	0/0 outside the five mechanisms: refutes "all are re

Note: the historical rows above carry the corpus's legacy 53-item tallies for the 55 former entries; the authoritative per-entry classification is the single-entry tables in Part II (entries 1-76).

Part IV: The Decision Tree

Given a suspected 0/0 form f/g at a point $x?$:

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Step 1: Verify  $f(x?) = g(x?) = 0$ 
  If not -> not a 0/0 form
  If yes -> proceed

Step 2: Compute the limit  $\lim_{x \rightarrow x?} f(x)/g(x)$ 
  If limit does not exist -> not a removable singularity
  If limit exists and is finite -> removable value found
  If limit is inf -> pole (not 0/0 in the useful sense)

Step 3: Classify the removable value
  Is it an integer?
    -> Yes: INDEX mechanism (topological invariant, multiplicity)
    -> No: proceed

  Is  $f/g = 1$  where defined (before taking the limit)?
    -> Yes: PROBE mechanism (tests functional equation)
    -> No: proceed

  Does the removable value depend on a rate of vanishing?
    -> Yes: VANISHING RATE mechanism (Taylor coefficient)
    -> No: proceed

  Is the 0/0 at a phase transition?
    -> Yes: CRITICAL PHENOMENON mechanism
    -> No: proceed

  Does the 0/0 arise from a symmetry?
    -> Yes: CONSERVATION mechanism
    -> No: Unknown mechanism -- classify it

Step 4: Verify computationally
  Compute  $f/g$  numerically near  $x?$ 
  Check convergence to the removable value

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Compare with the theorem's prediction

Part V: The Connections

5.1 How the mechanisms relate

The five mechanisms are not independent. They form a hierarchy:

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CONSERVATION <-?? creates ??-> PROBE
  ?                               ?
  ?                               ?
  ?                               ?
VANISHING RATE <-?? specializes ??-> INDEX
  ?
  ?
  ?
CRITICAL PHENOMENON

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Conservation creates Probe: A conserved quantity (Noether) implies a functional equation (zeta FE), which implies $f/g = 1$ where defined (the Probe). The Probe is the remnant of the conservation law.

Vanishing Rate specializes to Index: When the removable value is an integer, the Vanishing Rate mechanism specializes to the Index mechanism. The integer is the winding number, the multiplicity, or the topological invariant.

Critical Phenomenon is the physics of Vanishing Rate: At a phase transition, the order parameter and susceptibility both vanish (or diverge) at the critical point. Their ratio is a 0/0 whose removable value is the critical amplitude -- the universal constant that characterizes the universality class.

5.2 The unified picture

All five mechanisms share a common structure:

1. Two objects vanish at a point
2. Their ratio is 0/0
3. The removable value encodes structure

The difference is what kind of structure:

- Probe: structural identity (is $f = g$?)
- Index: topological invariant (what is the winding number?)
- Vanishing Rate: analytic invariant (what is the leading coefficient?)
- Critical: physical invariant (what is the critical amplitude?)
- Conservation: symmetry invariant (what is the conserved quantity?)

All five are invariants -- quantities that do not change under perturbation. The 0/0 form is the mechanism by which these invariants are extracted from the mathematics.

Part VI: Open Problems

6.1 Missing experiments -- computed 2026-09-20

Branch	Potential 0/0	Status
Galois theory	Discriminant of polynomial at repeated roots	Computed: $C = 32 / 27$ (Vanishing Rate)
Algebraic K-theory	Bott periodicity at degenerate spectra	Computed: period-2 ratio 4; degenerate value 1 (Index)
Geometric analysis	Ricci flow at neck pinch	Computed: $R \cdot A = 8\pi$ (Conservation)
Category theory	Natural transformation at degenerate objects	Computed: removable 1; $\dim 1 (k=1) / 2 = ?[S?] (k=2)$ (Probe)
Combinatorics	Generating function singularity at radius of convergence	Computed: amplitude $1/\sqrt{\pi}$, $\rho=1/4$ (Vanishing Rate)
Ergodic theory	Birkhoff average at exceptional points	Computed: removable 1 (generic) / 0 (exceptional) (Probe)
Functional analysis	Spectral density at band edge (general)	Computed: $\rho \cdot \sqrt{2E} \rightarrow 1/(2\pi)$ (Vanishing Rate)
Arithmetic geometry	Néron-Tate height at torsion points	Computed: log channel $C \sim 3/8$ (Vanishing Rate)

6.2 Classification completeness

Is the five-mechanism classification complete? We conjecture it is, based on the following argument:

Every 0/0 form f/g tests the relationship between f and g at $x?$. The possible relationships are:

1. $f = g$ (Probe)
2. f/g winds around $x?$ an integer number of times (Index)
3. f and g have specific leading coefficients (Vanishing Rate)
4. f and g diverge at a phase transition (Critical)
5. f and g are constrained by symmetry (Conservation)

We believe these five cover all possible relationships between two vanishing functions. A proof would require showing that every analytic 0/0 form falls into one of these categories.

Amendment (2026-09-20, entry 64). The strict form of that completeness claim is false: Pólya's $F(n)/\sqrt{n}$ is a 0/0 whose limit does not exist -- a genuine sixth relationship, " f/g has no limit" (the anti-class). Littlewood: $F = \Omega_{\pm}(\sqrt{n})$. The five mechanisms therefore classify every removable 0/0; the decision tree's "limit does not exist $\rightarrow$ not a removable singularity" branch is not empty, and conjectures that force a removable value onto such forms (Mertens, Pólya) are the framework's falsifiable targets.

Amendment (2026-09-20, entries 67-68 + detector). The anti-class now has a second member: Mertens $M(n)/\sqrt{n}$ (#67), whose limit Odlyzko-te Riele prove does not exist ($\limsup \text{ratio} > 1.06$, $\liminf < ?1.009$) and whose fluctuation exponent measures 0.433 (exact μ to $1e8$, $M(10^8) = 1928$). The fluctuation-scale criterion (affirmed P1-P5 in experiments/aclass_detector.py) now covers every open-target entry: Pólya beta = 0.486 (anti), PNT beta = 0.433 (removable), twin-prime beta = $?0.32$ (decaying to 1; carried in-entry to $1e8$ -- $\pi_2(1e8) = 440,312$, $R(1e8) = 0.9999$), Goldbach drift beta = $?0.09$ (no anti-class; in-entry decade maxima to $1e7$ decay beta = $?0.12$), prime-reciprocal beta = $?0.59$ (named constant).

6.3 The constructive problem

Can the 0/0 principle be used to discover new theorems? The approach would be:

1. Construct a 0/0 form f/g in a new setting
2. Compute the removable value numerically
3. If the removable value is "interesting" (an integer, a known constant, a simple expression), search for a theorem that explains it
4. If the removable value is "new" (not previously known), it may itself be a theorem

This is the constructive version of the 0/0 principle: not just verifying known theorems, but using 0/0 forms to find new ones.

6.4 The fluctuation-scale criterion and its falsifiable form

The anti-class is proved by five members (#64, #67, #69, #70, #74), so it calibrates the edge: for each candidate O/O with removable value c , let $D(x)$ be the per-decade max of

$R(x) \geq c$	(or, when sampling is too sparse, the pointwise log-log drift). The two readings of the criterion are the same statement in different units:
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- Ratio-deviation form (what the detector's P4 reads): $R(x) = f(x)/g(x)$,

removable-compatible $\geq D(x)$ falls decade-on-decade, so its log-log slope β_{ratio} is negative; anti-class $\geq D(x)$ does not converge to 0, $\beta_{\text{ratio}} \geq 0$.

- Fluctuation-size form (what P1/P2 read): if the leading scale of f is x^α

(Pólya $\sqrt[n]{n}$, PNT $x/\log x$), removable-compatible $\geq \beta_{\text{size}} < \alpha$, and the anti-class sits exactly on the denominator, $\beta_{\text{size}} \sim \alpha$.

The calibration is not a free choice: the six anti members have proofs pinning the failure point, so a return to the tree's "limit does not exist" branch needs a target whose decade maxima stop declining and floor strictly above 0 -- equivalently a sustained $\beta_{\text{ratio}} \geq 0$. Littlewood's $F = \Omega_{\pm}(\sqrt[n]{n})$ is exactly that statement (per-decade max of $F/\sqrt[n]{n}$ stays $\sim$ constant, $\beta_{\text{size}} \sim 0.5$, measured

0.486); Odlyzko-te Riele's $\limsup M/\sqrt[n]{n} > 1.06$ does the same for Mertens (measured exponent 0.433); Montgomery-Vaughan Thm 15.11's $\psi \geq x = \Omega_{\pm}(x^{1/2})$ does the same for Chebyshev (measured band 0.51-0.61-0.71-0.69-0.68-0.75, sup 0.74 @ 36.9M -- below the $\sqrt[n]{n}$ floor, so the finite band alone would be removable-looking; the theorem, not the range, is the arbiter); Paley's $R(q) \geq c \log \log q$ does it in q -space (measured floor 0.577+). On the removable side the decay signs observed across the corpus all lie below the anti floor: prime-reciprocal decade maxima

0.605 $\rightarrow$ 2.7e25 ($\beta_{\text{ratio}} = ?0.59$), twin-prime 0.0198 $\rightarrow$ 0.0096 $\rightarrow$ 0.0046 ($\beta_{\text{ratio}} = ?0.32$, full decade to 1e8), and PNT's ratio deviation $(\pi(L)/L) \sim x^{?0.57}$ -- the fluctuation is subleading to x , removable -- and the totient density (#71) collapses with $\beta_{\text{ratio}} = ?1.20$: its error is a whole power below its x^2 denominator, the removable side's fastest legitimate decay.

Entry #69 (the Dirichlet divisor-error, Hardy 1916) adds a stronger member and one structural lesson. $\Delta(x)/x^{1/4}$ is proven to have no limit -- the $\Omega_{\pm}((x \log x)^{1/4} \log \log x)$ theorem makes its $\limsup +\infty$ unconditionally, the $\Omega_{\pm}(x^{1/4})$ pattern gives infinitely many negative values -- so no finite wall can display the failure, and the decade-max ratio series reads order-one (3.4 $\rightarrow$ 3.8 $\rightarrow$ 4.8 $\rightarrow$ 5.4 $\rightarrow$ 5.8 $\rightarrow$ 1.7, $d = 2..7$ to $1e7$) with a single near-threshold dip ($\Delta(10^7) = ?4$, $S(10^7) = 162,725,364$) rather than a clean slope. The lesson: for members whose oscillation provably lives on the denominator, the bounded-band reading -- the per-decade max of the normalized deviation never collapsing below order-one -- is the robust anti signature, and a single decade whose deviation cancels near a round point must not be read as decay. Removable members, by contrast, collapse their band by orders of magnitude per decade (twin 0.0198 $\rightarrow$ 0.0046; prime-reciprocal 0.605 $\rightarrow$ 2.7e25; PNT $\beta_{\text{ratio}} \sim ?0.57$).

Entry #70 (the Gauss circle-error) is the Hardy-pair twin on the SAME $x^{1/4}$ denominator with the OPPOSITE unbounded side ($\liminf = ?\infty$), and its numerical profile is the same bounded-band picture (per-decade max ratio

4.9 $\rightarrow$ 5.4 $\rightarrow$ 6.2 $\rightarrow$ 7.3 $\rightarrow$ 8.0 $\rightarrow$ 1.75 to $1e7$, with the same near-threshold dip at $P(10^7) = 98$). Together #69/#70 show the lattice-point anti-class is not an accident of one construction: the denominator-carrying fluctuation is the shared, theorem-pinned structure of an entire family.

Why, then, is Goldbach's gate the weak form ("no growth $\geq +1/2$ ") and not the strong form (" $\beta_{\text{ratio}} < ?0.2$ ")? Two structural reasons that the criterion admits without weakening:

1. Range. [1e4, 3e5] spans two full decades, and the per-even wobble --

mean $R \geq 1 \sim 0.2$, worst sample 0.34 -- swamps any log-slow decay, so a

tight negative slope is unmeasurable at this reach. A positive anti-class drift, by contrast, would be plainly visible (decade maxima climbing above

0.34). The measurable distinction at this range is growth versus no-growth, which is why G6 reads the drift (?0.09) instead of a decay threshold.

2. What the weight already removed. S(N) strips the smooth first-order

dependence: $G(N)/(S(N) \cdot N/\log^2 N)$ is flat to leading order, and entry 66's Spearman window ($\rho > 0.6$ within a fixed N-window) shows the residual is exactly the singular-series ordering -- a bounded, zero-mean process. For a bounded process the only way to threaten removability is to grow, so the criterion's dangerous clause -- a validated run whose $R_{?1}$ climbs, decade maxima flooring above a positive constant -- is the strictest clause the range can legally assert.

The same Criterion therefore sharpens as range increases: demand a decay sign where full decades are sealed (twin at $1e8$, prime-reciprocal at $2e7$, Mertens and Pólya at $1e8$), demand only a non-growth where reach is two decades (Goldbach at $3e5$ in the detector, P5). Range extension delivers exactly that sharpening: in-entry at $1e7$ (260 log-uniform evens) the Goldbach decade maxima are $0.296 \rightarrow 0.226 \rightarrow 0.174$, decade-decay $\beta = ?0.12$ ($r^2 = 1.00$, three decades) -- the shallow decay the two-decade reach could not measure -- while growth (decade maxima flooring/climbing above the bounded band) remains the operationally falsifiable clause. A later decade of Goldbach counts that flips the criterion's decay gate to FAIL is the ledger's PL-11 refutation, and it is a stronger falsification than a mere slope miss: it pushes the target onto the proved anti-class.

6.5 The proved-anti calibration: finite ranges always "look removable"

The six anti-class members are the corpus's calibration set, and what they calibrate is uncomfortable: in every finite verification an anti member is

indistinguishable from a removable one. No range has ever shown any of them failing the bound it was designed to violate.

- Pólya (#64): $F(n) \leq 0$ held for $> 9 \times 10^7$ terms before the first + at

$n = 906,150,257$; Littlewood's $\Omega_{\pm}$ guarantees infinitely more violations. The O/O-limit for $F(n)/\sqrt{n}$ failed only in the theorem, never in the data.

- Mertens (#67): verified here to $1e8$ -- the near-miss ratio $M(n)/\sqrt{n}$

grows with the range (0.4378 @ $300,551 \rightarrow 0.4627$ @ $30,919,091$), the bound HELD on $[2, 1e8]$ (max ratio 0.8944 , and that at $n = 5$), records still being set at $n = 76,015,339$, and the conjecture is only proved false (Odlyzko-te Riele, $\limsup > 1.06$) -- every finite value "looks removable".

- Divisor-error (#69): the sharpest case. Verified here to $1e7$ -- $\Delta/x^{1/4}$

stays order-one (~ 1.7 - 5.8) in every decade with signs oscillating, and

Delta	$ x^{1/3} \leq 1.93$ (Voronoi's 1/3 bound). Because Hardy's $\Omega_{\pm}$ theorem
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the ratio's $\limsup +\infty$ unconditionally, no finite range can ever display the failure -- this member's verdict rides entirely on the theorem, and the finite data can only corroborate the structural reading (order-one band, oscillation), never exhibit the contradiction.

- Circle-error (#70): the twin. $P(x)/x^{1/4}$ reads the same bounded-band

profile ($4.9 \rightarrow 8.0 \rightarrow 1.75$ per-decade max to $1e7$, $P(10^7) = 98$) on the opposite unbounded side (Hardy's $\liminf = ?\infty$); the lattice-point pair shows the anti-class is a family structure, not a construction accident.

- Equal-range wall (10^8): both lattice members re-verified exactly to

$1e8$ via their closed forms ($S(x) = \sum x/d$, $N(x) = (4m+1)+4\sum \sqrt{x-a^2}$ -- identities certified by the #69/#70 sieves to $1e7$; continuity to the wall). Sampled-window maxima keep rising, not decaying -- divisor

$2.568 \rightarrow 2.643 \rightarrow 2.913 \rightarrow 2.992$, circle $4.243 \rightarrow 3.106 \rightarrow 4.420 \rightarrow 4.599$ (full floors $2.57 / 3.11$) -- and the near-threshold 10^7 cancellation is confirmed as a dip inside the band, not a trend: window $10^7..10^8$ beats window $10^6..10^7$ for both. $S(10^8) = 1,857,511,568$, $N(10^8) = 314,159,053$ ($\Delta/x^{1/4} = 3.61$, $P/x^{1/4} = 2.12$ at the wall). The lattice pair now holds the same $1e8$ range as the $\sqrt{n}$ pair (#64/#67).

- sqrtn pair at the wall (10^8): Pólya #64 extended to $1e8$ alongside

Mertens #67 for the pairwise calibration -- per-decade band max

F	/sqrtn = 1.29	1.33	1.30	1.27	1.36 (floor 1.27, never decaying),
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family amplitude already above 1 at $1e7$ (1.167 @ 8,803,471) and max

1.358, while $M/\sqrt{n}$ stays inside the conjectured bound at range (0.8944). $F(10^k) = 0, ?2, ?14, ?94, ?288, ?530, ?842, ?3,884$ (A002819; $a(10^8) = ?3884$), min $F = ?10,443$ @ 76,015,169 -- the deepest excursion sits mid-band, and $F(10^8) = ?3,884$ is the same *single-point pull-back* structure the lattice pair shows at 10^7 (Delta, P) and #72 at its wall: a band truth (1.358) that a naive endpoint read (0.388) would misreport as decay -- the third independent occurrence of the Dip Rule (§6.7).

- Chebyshev completes the sqrtn trinity (#74): $\delta(x)/\sqrt{x}$ at $1e8$, $\delta = \psi(x) ? x$,

computed exactly via the prime-power breakpoints of ψ (= log lcm($1..?x?$), spot-verified to $1e9$ on 1..40). $\psi(10^8) = 99,998,242.8$ (ratio 0.999982), $\delta(10^k) = ?2.2, ?6.0, ?3.3, +13.4, +51.6, ?413.4, ?1460.6, ?1757.2$ -- both signs, crossing the axis afresh in every decade. Per-decade band max

delta	/sqrtx ($d = 2..7$) = 0.51 · 0.61 · 0.71 · 0.69 · 0.68 · 0.75 (non-collapse: each band holds $\geq ?$ of the previous, through the wall), sup 0.74 @ 36,917,099 -- a reader between the sqrtn pair's floors ($0.46 < 0.75 < 1.27$), which is the member's whole point: Montgomery-Vaughan Thm 15.11 ($\psi ? x = \Omega_{\pm}(x^{1/2})$, unconditional), not any finite read, is what makes delta anti. It is the corpus's clearest proof-before-range exhibition -- the mirror of #69/#70, where the finite read looked decaying and the theorem still decided.
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The wall in one read (all five anti members at the 10^8 wall):

member	window-max read (same as window-max for all)	wall point	verdict
Mertens #67 (sqrtn)	M	$\sqrt{n} = 0.472$ @ 2,803 ($n \geq 10^3$)	sqrtn @ 894,910,091 (Dip Rule) band flat in
Pólya #64 (sqrtn)	per-decade 1.29 · 1.33 · 1.301 @ 23, ?2.36, floor $F(10^8) = ?3,884$: point 0.388 (Dip)		amplitude already above
Chebyshev #74 (sqrtn)	per-decade ($d = 2..7$) = 0.51 · 0.61 · 0.71 · 0.69 · 0.68 · 0.75 (non-collapse)		band real
Divisor #69 ($x^{1/4}$)	windows 2.57 · 2.64 · 2.61 · 2.59 (for 2..57): 3.38 · 3.84 (4.03) 5.38 / 1002 = 1.68 (a rise)		limit unconditionally (Hardy); the ho
Circle #70 ($x^{1/4}$)	windows 4.24 · 3.14 · 4.22 · 4.60 (for 2..7): 4.89 · 5.41 (10^8) 21 · 7.28 / 1002 = 2.72 (rise)		limit unconditionally (Hardy); its dip

The sqrtn trinity ($F 1.27 \cdot M 0.46 \cdot \delta 0.75$ late bands) and the $x^{1/4}$ pair (2.57 / 3.11) all read the same in-band non-decay at the same wall -- size never decides (0.46 sits below 0.75 sits below 1.27 sits below 3.11), the band's flatness does: which is the criterion's own claim, and exactly what the Omega-theorems render a true limit failure rather than a finite cap.

Consequences for the criterion:

1. A near-miss ratio below 1 is evidence of nothing by itself: Pólya and

Mertens both sit strictly below 1 in every range that exists to compute, and the divisor/circle ratios, though already order-one, are capped by any finite wall exactly when Hardy's theorems say the failure is a limit one has to reason to, not a value one reaches.

2. The only legally assertable anti-signature is non-decay -- decade maxima

flooring strictly above 0, a floor the proofs pin on the Mertens bound ($\limsup > 1.06$, $\liminf < ?1.009$) and declare unbounded-above and unbounded-below for the lattice-point pair and for delta and F by their own $\Omega_{\pm}$ theorems -- and §6.4's falsifiable form for open targets is exactly the complementary clause: a validated run whose decade maxima stop declining and floor above a positive constant, while the "bound" keeps holding only empirically.

3. So a target that does decay decade-on-decade (prime-reciprocal ?0.59,

twin ?0.32, totient ?1.20, character-box ?0.59) is exhibiting a behavior the six anti members provably never exhibit -- which is the full content of "removable-compatible": support, never proof. The criterion separates the corpus into the two classes by a signature the anti class provably lacks, and concedes that no finite run can prove removability of an open target. (Entry #72, a manufactured control on the removable side -- the same character walk built to never settle, whose

bounded box puts its measured slope at ~ 0.59 -- pins item 3's converse: the criterion grants a bounded-band raw deviation the removable read precisely when the corrected ratio collapses, and the anti members are the case where it provably cannot. Entry #73 shows the other face of the same family: release the box -- let the modulus grow -- and the same character walk's sup-ratio provably stops converging, Paley's theorem.)

6.6 The three structural readings and the bounded-box control

The whole calibration rests on ONE observable: the log-log slope β_{ratio} of the per-decade maxima of deviation / denominator. The corpus now rules it in three regimes, with members whose verdicts are proved on both sides:

- Anti, order-one band ($\beta_{\text{ratio}} \sim 0$, floor > 0): Pólya 0.486, Mertens

0.433 (sqrt denominator, limsup pinned 1.06 / 1.009), and now the Chebyshev walk $\delta/\sqrt{x}$ (#74: per-decade bands $0.51 \cdot 0.61 \cdot 0.71 \cdot 0.69 \cdot$

$0.68 \cdot 0.75$, sup 0.74 @ 36.9M -- a reader BELOW the sqrt pair's floors, which is the point: Montgomery-Vaughan Thm 15.11, not any range, is what makes δ anti), divisor/circle ($x^{1/4}$ denominator, band 1.7-8.0 vs 4.9-8.0, no limit unconditionally), character-walk in q -space (#73: floor 0.69, creeping sup, no limit by Paley). The class is nearly closed within each space: the classic n -space summatory functions are now the sqrt trinity ($F = \text{Sigmalambda}$, $M = \text{Sigmamu}$, $\delta = \psi \cdot x$ -- all three $\Omega_{\pm}(\sqrt{x})$) plus the $x^{1/4}$ lattice pair -- the verdicts proven on both sides -- while the two classics still OPEN at the bare denominator are $\pi?Li$ and $S(T)$: both are proven to oscillate (Littlewood 1914: $\pi?Li = \Omega_{\pm}(\sqrt{x} \cdot \log \log x / \log x)$; Selberg: $S(T)$ unbounded both signs) and both sit at scales a log (or less) BELOW their naked denominators, but no theorem proves the ratio's tendency to zero -- the best upper bounds (even under RH: $\pi?Li = O(\sqrt{x} \log x)$, $S(T) = O(\log T)$) leave the ratio room up to $\log x$ -- so neither is proved decaying, nor proved anti. What provably decays in n -space is the bounded-box family: fixed-modulus character partial sums (#72, period-mean-zero $? S(x) \leq c \cdot \text{ratio} \rightarrow 0$, Pólya-Vinogradov). The q -space move (Paley) and the lattice-power move (Hardy) each open a new anti direction the criterion's seamless band-read confirms.

- Removable, collapsing ratio ($\beta_{\text{ratio}} < 0$ strictly): prime-reciprocal

0.59 , twin 0.32 , totient 1.20 .

- Removable, bounded-box (the manufactured control, #72): the fixed

quadratic character mod 7 -- $S(x) \leq 2$ for every x (one-period max, box ratio 1.00 to $1e7$) yet $S(x)/\sqrt{x} \rightarrow 0$ at measured $\beta_{\text{ratio}} = 0.590$ ($r^2 =$

0.985 , the expected $1/2$): a case where the raw deviation never settles and never grows, and the corrected ratio still provably collapses. The control answers the criterion's border question -- "what if the deviation merely wiggles in a bounded band forever?" -- with Pólya-Vinogradov: for fixed moduli that band is a box and the read is removable; the Paley oscillation ($\Omega_?(\sqrt{q} \log \log q)$) lives across moduli, a different O/O in q -space the criterion's denominator does not address. The Chebyshev-bias ($\pi(x;4,3) - \pi(x;4,1) \sim 1.79\sqrt{x}/\log x$) is the honest intermediate: a band that does grow, one log below $\sqrt{x}$, so the corrected ratio (band/denominator) decays -- removable, at rate slower than any boxed control, which is exactly why the slope read, not the band's size, is the criterion.

Collectively: fluctuation size is never evidence; the ratio's decade trend is the whole observable; and the anti class is thin, provable, and nearly exhaustive, while the removable side is measure-theoretically sweeping -- the asymmetry §5.2 predicted as Principle 3 in §4.4.

6.7 The Dip Rule

A single-point cancellation at a power-of-ten wall never defines a trend; only window (band) comparisons do. The corpus has now hit the same near-threshold structure three times, independently, and reading the point instead of the band would have falsified a true anti member each time:

- $\Delta(10^7) = ?4 \rightarrow$ point ratio 1.68 vs. the $[10^6, 10^7]$ window's 5.80

(divisor #69, extended):

- $P(10^7) = 98 \rightarrow$ point 1.75 vs. window 8.04 (circle #70, extended);
- $F(10^8) = ?3,884 \rightarrow$ point ratio 0.388 vs. the $[10^7, 10^8]$ window's

1.358 (Pólya #64, extended to $1e8$) -- the deepest excursion sits mid-band ($?10,443 @ 76,015,169$), the walk then pulls back through the wall.

Every one of these is a genuine cancellation of the fluctuation AT the decade point -- the exact feature a decay-sign read on endpoints would confuse with convergence. The rule is therefore the criterion's second half, complementing §6.6's band reads: ****the observable is the window maximum; the decade point is decoration.**** The three occurrences are the same phenomenon across three different arithmetic families (sqrtn, $x^{1/4}$, and the character/lattice family), which is why the extension probes (#64 @ $1e8$, #69/#70 @ $1e8$) were built to sample windows first and report landmarks second.

6.8 The three sqrtn walks read the same slope, different bands

The criterion's slope reads the CLASS, not the member -- shown by the sqrtn trinity measured pairwise at the $1e8$ wall:

walk	per-decade band max ratio	sup	wall point
Pólya F/sqrtn (#64)	$1.29 \cdot 1.33 \cdot 1.30 \cdot 1.27 \cdot 1.36$ (d = 7)	1.358 (d = 7)	$F(10^8) = ?3,884: 0.388$ (dip)
Mertens M/sqrtn (#67)	flat ~ 0.46 (max $\geq 1e5$: 0.4630463046, d = 7)	0.463 (d = 7)	$M(10^8) = +1,928: 0.1928$ (dip)
Chebyshev delta/sqrtn (#74)	$0.51 \cdot 0.61 \cdot 0.71 \cdot 0.69 \cdot 0.68$ (d = 7)	0.745 (d = 7)	$\text{delta}(10^8) = ?1,757: 0.176$ (dip)

All three read log-log slope beta_ratio ~ 0 -- flat bands -- because all three are governed by the SAME mechanism: the partial sums of (Möbius, Liouville, Chebyshev-psi) arithmetic functions, whose deviations are all $\Omega_{\pm}(\sqrt{x})$ with sqrtx exactly the natural scale, the correction locked by the Riemann-zero machinery. The slope cannot separate them; the band SIZE is the family fingerprint (F's floor 1.27 -- least cancellation; M's 0.46 -- most; delta's

0.75 -- the prime-power log-sum between). And their three wall POINTS are the Dip Rule trinity: each 10^8 read sits at a minimum of its own window (0.388 vs 1.358; 0.1928 vs 0.463; 0.176 vs 0.748) -- endpoint reads would have called all three "decaying". The verdicts ride the theorems, exactly as §6.5's arbiter row states: the pair's bands are inside $1.27 \leftrightarrow 0.46$, delta's is between them.

6.9 The OPEN pair (fenced, uncertified): $\pi?Li$ and $S(T)$

The criterion's slope is necessary, not sufficient: an order-one band can mean "proven anti" OR "open, fluctuating below the denominator". The two classic n-space sums left over by §6.6 sit in the second slot, and neither side can certify them -- the one computed reading (exact scan 2026-09-21, primes through $R(x) = 10^8$ sieved exactly and checked against the known $\pi(10^k)$ line 4, 25, 168, 1229, 9592, 78498, 664579, 5761455; Li to $< 10^{23}$ via the convergent Ei series):

member	O/O form	ratio at 10^k ($k = 1..8$)	reading to $1e8$	theorem side	status
$\pi?Li$ open classic	$(\pi(x) ? Li(x))/\sqrt{x}$	0.685, 0.513, 0.304, 0.171, 0.129, 0.130, 0.130, 0.130	0.130 (dip)	Li: $\pi(x) ? 2x/\ln x$	OPEN
$S(T)$ open classic	$S(T)/\log T$, $S(T) = (1/\pi) \arg \zeta(1/2 + it)$			Selberg: $S(T)$ unbounded	OPEN

Two structural facts the pair exhibits that the certified classes never do:

1. ****The decade point is the window maximum, not the dip (the Dip Rule's**

inverse).** For $\pi?Li$ the gap-lemma holds: $g'(x) < 0$ in every gap (the stationary equation $Li(x) ? \pi = 2x/\ln x$ is never met -- at $1e8$ the left side is 754, the right ~ $10^?$), so within a band $g = (Li/\pi)/\sqrt{x}$ rises RIGHTWARD and its sup lands ON the decade point. The certified anti trio's sup sits MID-window, its decade point a minimum (Dip Rule, §6.7). The open family flips it: the single-sign walk climbs through the wall instead of pulling back through it.

2. The slope reading hover-strikes its own soft negative (~ $?0.07$).

No certified member reads there: the removable class is sharply negative ($?0.32, ?0.59, ?1.20$) and the anti class is flat (~ 0). $\pi?Li$'s intermediate, direction-shuffling beta is exactly the lurking Littlewood factor $\log \log \log x / \log x$ leaking through the range -- a finite "wink" of decay that nobody has promoted to a theorem. The fence therefore marks the

boundary where the framework declines to classify: without a certified value (removable) or a negative oscillation theorem (anti), a 0/0 stays open, no matter how believable its finite trend reads. No census entry is claimed (§6.9 is a fence, not a member): the 6 anti members and the 91 removable-value entries stand, and the open pair is what the framework can point to and say -- this is where it must wait.

Appendix: Data Files

Each experiment produces a JSON data file in data/. The files contain:

- The 0/0 form and its parameters
- Numerical values of the removable value at multiple points
- Convergence data (how the removable value is approached)
- The summary verdict (SUPPORTED / NOT SUPPORTED)
- Honest wall statements (limitations of the numerical verification)

Experiment	Data File
Riemann zeta	data/argument_principle_0_over_0_data.json
GRH Dirichlet	data/grh_dirichlet_0_over_0_data.json
BSD	data/bsd_0_over_0_data.json
abc	data/abc_conjecture_0_over_0_data.json
Fermat little	data/fermat_little_0_over_0_data.json
Euler product	data/euler_product_0_over_0_data.json
Weil explicit	data/weil_explicit_0_over_0_data.json
Zeta FE	data/zeta_functional_eq_0_over_0_data.json
PNT	data/prime_number_theorem_0_over_0_data.json
Khintchine	data/khintchine_0_over_0_data.json
Möbius	data/mobius_function_0_over_0_data.json
Argument principle	data/argument_principle_0_over_0_data.json
Cauchy integral	data/cauchy_integral_0_over_0_data.json
Picard	data/picard_little_0_over_0_data.json
Taylor	data/taylor_remainder_0_over_0_data.json
FTA	data/fta_0_over_0_data.json
Stirling	data/stirling_approx_0_over_0_data.json
Wallis	data/wallis_product_0_over_0_data.json
Cesàro	data/cesaro_summation_0_over_0_data.json
Poincaré-Hopf	data/poincare_hopf_0_over_0_data.json
Atiyah-Singer	data/atiyah_singer_0_over_0_data.json
Gauss-Bonnet	data/gauss_bonnet_0_over_0_data.json
Riemann-Roch	data/riemann_roch_0_over_0_data.json
Weyl law	data/weyl_law_0_over_0_data.json
Selberg trace	data/selberg_trace_0_over_0_data.json
Lefschetz	data/lefschetz_fixed_point_0_over_0_data.json
Morse theory	data/morse_theory_0_over_0_data.json
Sard	data/sard_theorem_0_over_0_data.json
Stokes/de Rham	data/stokes_de_rham_0_over_0_data.json
Green's function	data/greens_function_0_over_0_data.json
Euler-Maclaurin	data/euler_maclaurin_0_over_0_data.json
CLT	data/central_limit_theorem_0_over_0_data.json
Rayleigh	data/rayleigh_quotient_0_over_0_data.json

Banach	data/banach_fixed_point_0_over_0_data.json
Brouwer	data/brouwer_fixed_point_0_over_0_data.json
Fourier uncertainty	data/fourier_uncertainty_0_over_0_data.json
Poisson summation	data/poisson_summation_0_over_0_data.json
Saddle point	data/saddle_point_0_over_0_data.json
Laplace	data/laplace_method_0_over_0_data.json
Noether-Landau	data/noether_landau_0_over_0_data.json
Ising	data/ising_model_0_over_0_data.json
Spectral gap	data/spectral_gap_0_over_0_data.json
Lorenz	data/lorenz_attractor_0_over_0_data.json
Wigner	data/wigner_semicircle_0_over_0_data.json
Shannon	data/shannon_entropy_0_over_0_data.json
Boltzmann	data/boltzmann_entropy_0_over_0_data.json
Bayes	data/bayes_theorem_0_over_0_data.json
Gradient descent	data/gradient_descent_0_over_0_data.json
KKT	data/kkt_conditions_0_over_0_data.json
Schanuel	data/schanuel_0_over_0_data.json
Noether	data/noether_theorem_0_over_0_data.json
Pythagorean	data/pythagorean_0_over_0_data.json
Pólya anti-class	data/polya_liouville_0_over_0.json
Galois discriminant	data/galois_discriminant_0_over_0.json
Bott periodicity	data/bott_periodicity_0_over_0.json
Ricci flow neck pinch	data/ricci_flow_neck_pinch_0_over_0.json
Natural transformation	data/natural_transformation_0_over_0.json
Generating function singularity	data/generating_function_singularity_0_over_0.json
Birkhoff average	data/birkhoff_average_0_over_0.json
Spectral density (band edge)	data/spectral_density_band_edge_0_over_0.json
Neron-Tate height (torsion)	data/neron_tate_height_torsion_0_over_0.json
Twin-prime density	data/twin_prime_density_0_over_0.json
Goldbach density	data/goldbach_density_0_over_0.json
Mertens function	data/mobius_mertens_0_over_0.json
Prime-reciprocal constant	data/prime_reciprocal_0_over_0.json
Dirichlet divisor-error (anti-class)	data/dirichlet_divisor_anti_0_over_0.json
Gauss circle-error (anti-class)	data/gauss_circle_anti_0_over_0.json
Euler-totient density	data/euler_totient_0_over_0.json
Character partial sums (bounded-box control)	data/character_sums_0_over_0.json
Character-walk sup-ratio (anti-class, q-space)	data/char_walk_anti_0_over_0.json
Lattice pair at the 10^8 wall (extension of #69/#70)	data/lattice_wall_1e8_0_over_0.json
Pólya/Liouville walk at the 10^8 wall (extension of #64)	data/polya_wall_1e8_0_over_0.json
Chebyshev walk $\delta/\sqrt{x}$ (anti-class)	data/chebyshev_psi_0_over_0.json
Mertens product complement	data/mertens_product_0_over_0.json
Fluctuation-scale detector (aclass)	data/aclass_detector.json
Landau function	data/landau_function_0_over_0_data.json

The Landau-function calibration (corpus #56, PL-21; registered 2026-09-22) -- REMOVABLE named value 1, Vanishing Rate, Landau 1903: $\log g(n)/\sqrt{n \log n} \rightarrow 1$ unconditionally (A000793, 5/5 gates PASS, settled referee) -- is the numbered census's 76th entry (§2.20, above).

Cross-Domain Applications (physics fields using the Index mechanism)

Mechanism II (Index: winding number; THE_0_OVER_0_ATLAS.md:35) is not only a mathematical taxonomy -- the framework's own physical lanes run on it:

Physical framework	O/O mechanism	Artifact	Status
FRG pole-pair winding: lower ridge w/ winding number $\nu=0$ (exp. exp. part of winding) regular	INDEX (winding number) and a positive residue (the helical selection)	ON_CROSSING (experiments) winding (A-B) diagram	
Release-lane selection: the CMB (folded) and a positive residue (the helical selection)	ON_CROSSING (experiments) winding (A-B) diagram	ON_CROSSING (experiments) winding (A-B) diagram	
Higgs handoff: ridge eject lam	INDEX (winding number) and a positive residue (the helical selection)	ON_CROSSING (experiments) winding (A-B) diagram	
Stability of the ejector under resonance	INDEX (winding number) and a positive residue (the helical selection)	ON_CROSSING (experiments) winding (A-B) diagram	
Regulator-resolution	INDEX (winding number) and a positive residue (the helical selection)	ON_CROSSING (experiments) winding (A-B) diagram	

Cross-reference: AUDIT.md (Poincare-Hopf index = $\chi(M)$; argument-principle zero counts), WHERE_0_OVER_0_SOLVES.md, THE_UNIVERSAL_ZERO.md §2.2 (Index mechanism). The winding number in the physics lane and the residue-theorem family above are the same object: an integer removable value of a 0/0.

This atlas is a reference document. For the philosophical interpretation, see ON_THE_NATURE_OF_ZERO.md. For the synthesis, see THE_UNIVERSAL_ZERO.md. For the epistemology, see REMOVABLE_SINGULARITIES.md.

All 91 removable-value entries verified computationally (85-sweep + 6 open-target/named/control -- the sweep's 64th, the Landau-function calibration #76 -- REMOVABLE value 1, Vanishing Rate, Landau 1903; the sixth, the Mertens product-gap complement #75 -- REMOVABLE value 0, elementary, Mertens 1874; the sweep's 65th, the Crossing-a-zero calibration (corpus census #80) -- REMOVABLE value 0 ($j \geq 1$) / v_0 ($j = 0$), Critical Phenomenon, and its parity law: at a mutual zero of gap j the closed-form ratio $(x^{?1})^j \cdot V/U$ either carries the value $v_0 = V(1)/U(1)$ across ($j = 0$) or arrives at rate $v_0 \cdot x^{?1 \wedge j}$ ($j \geq 1$), crossing zero iff j is odd -- verified exact at 60 digits, two-sided, with certified rationals 2048/65537 and 65537/67108865 as value and rate); and the sweep's closing 66th-85th, the 20-experiment round-out (corpus census #81-#100) -- the Table of Great Limits (sinc, exponential rate, continuity of e , half-cosine, Taylor third, polygon perimeter, $n^{(1/n)}$, Euler-Mascheroni, factorial root, central binomial), the arithmetic-landscape probes (Dirichlet divisor-summatory, sigma-summatory, primes-in-AP, consecutive-prime ratio, partition, Li-form prime counting, squarefree density, Gauss-circle), the Newton $1/(2\sqrt{2})$ quadratic rate, and the golden-ratio Binet collapse -- all removable values, all passed; entries 64, 67, 69, 70, 73 and 74 document the anti-class -- the Pólya and Mertens conjectures are FALSE, proven externally (Littlewood; Odlyzko-te Riele), the Chebyshev-walk ratio has no limit, proven unconditionally (Montgomery-Vaughan Thm 15.11: $\psi(x) = O(x^{1/2})$), the divisor/circle ratios have no limit, proven unconditionally (Hardy 1916; Voronoi; Sierpiński), and the character-walk sup-ratio has no limit in q -space, proven unconditionally (Paley 1932; Pólya-Vinogradov 1918) -- each borne out by its fluctuation reading (Pólya 0.486, Mertens 0.433, Chebyshev delta band 0.51-0.61-0.71-0.69-0.68-0.75 below the sqrt floor -- the theorem, not the range, arbitrates, divisor/circle bounded-band order-one, character-walk floor 0.577+ with creeping sup); entry 71 documents the removable side's flat-decay anchor ($3/\pi^2$, beta_ratio = ≈ 1.20 , Dirichlet 1849); entry 72 documents the bounded-box control (mod-7 character partial sums: $S \leq 2$ at every x to $1e7$, $S(x)/\sqrt{x} \rightarrow 0$ at beta_ratio = ≈ 0.590 , Pólya-Vinogradov) -- the manufactured counterpoint proving the criterion reads the ratio's decade trend alone, never the size of the raw oscillation. 690 regression tests passing.